

## Research papers

# Changes in preferential flow caused by root effects in black locust plantations of different stand ages in the semi-arid region of the Loess Plateau

Lu Cai<sup>a</sup>, Fei Wang<sup>a,b,\*</sup>, Yuanyuan Lin<sup>a</sup>, Qi Long<sup>a</sup>, Yulong Zhao<sup>a</sup>, Jianqiao Han<sup>a,b</sup>, Wenyan Ge<sup>a,b</sup>, Hao Chen<sup>a,b</sup>

<sup>a</sup> Institute of Soil and Water Conservation, Northwest A&F University, Yangling, Shaanxi, China

<sup>b</sup> Institute of Soil and Water Conservation, Chinese Academy of Science and Ministry of Water Resources, Yangling, Shaanxi, China

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## ABSTRACT

Preferential flow significantly influences regional water cycles, and affecting surface and subsurface runoff processes. By comparing soil infiltration characteristics across various forest ages, it can be evaluated how artificial black locust forests on the Loess Plateau contribute to the resilience of soil and water conservation. In this study, we analyzed the spatial distribution characteristics of roots of black locust plants using field investigations and statistical analyses and determined the effects of the root characteristics on the preferential flow in the semi-arid of the Loess Plateau, China. The results showed that the root biomass, root length density, and root volume ratio of young forests were significantly lower than those of middle-aged and mature forests ( $P < 0.05$ ), and root characteristic parameters varied significantly among different root diameters and soil layers. The average volume of decayed root channels in mature forests, accounting for approximately 0.16 % in a unit volume of soil, was 3.1 times that of young forests and 4.9 times that of middle-aged forests. Preferential flow increased with forest age in artificial black locust forests. 2–5 mm, 30–40 mm and > 50 mm roots promoted preferential flow to varying degrees, while roots of 5–15 mm inhibited the preferential flow. < 2 mm, 10–30 mm and 40–50 mm roots has no effect on preferential flow ( $P > 0.05$ ). The size and density of decayed root channels significantly improve soil preferential flow, and the differences among forest ages are linked to the network created by rotten roots. These findings indicate that living roots of different diameters and decayed roots contribute to soil preferential flow to varying extents. The results provided a conceptual basis for understanding the potential impact of vegetation roots on surface runoff during the vegetation restoration ecosystems process in the Loess Plateau.

## 1. Introduction

Preferential flow refers to the non-equilibrium flow process in which water and solutes in the soil rapidly migrate to the deep layer along the preferential path in the soil. Preferential flow accounts for approximately 11 %–85 % of the total infiltration volume (Stone and Wilson, 2006, Legout et al., 2009, Laine-Kaulio et al., 2014, Zhang et al., 2021), which has a very important impact on the regional water cycle (Chen et al., 2021, Zhao et al., 2023). Preferential flow affects surface and subsurface runoff processes, and promotes soil moisture recharge. The Loess Plateau is one of the most fragile ecosystems in China. The vegetation restoration model constructed by the Grain for Green Project has

developed from the initial simple artificially planted vegetation system to a complex artificially planted vegetation system that can reverse desertification. However, exclusively concentrating on the ecological outcomes of restoration is inadequate for assessing the overall success of restoration projects (Wortley et al., 2013, Brancalion and Holl, 2020, Luo et al., 2019a). The imbalance in the forest ecosystem between water supply and demand has become notably severe, emerging as a central concern in ecological conservation and management (Wang et al., 2011, Jia and Shao, 2014, Jian et al., 2015, Huang and Shao, 2019). The challenge of deep soil drought poses a constraint on the advancement of artificial forests and hinders the advancement of ecological restoration efforts (Wang et al., 2010, Jia and Shao, 2014, Liu et al., 2018b).

\* Corresponding author at: No.3 Taicheng Road, Institute of Soil and Water Conservation, Northwest A&F University, Yangling, Shaanxi Province, China.  
E-mail address: [wafe@ms.iswc.ac.cn](mailto:wafe@ms.iswc.ac.cn) (F. Wang).

Conducting research on preferential flow in the Loess Plateau region is crucial for improving water resource utilization efficiency, comprehensively evaluating soil and water conservation benefits, slowing down the degradation of artificial vegetation, and fostering sustainable environmental practices (Kulasekera et al., 2011).

Forest soil structure has significant spatial variability, riching in large pores and quantity due to the smaller disturbance of humans. Preferential flow type is mainly large pore flow. According to whether the formation of soil macropores is affected by biological effects, it can be divided into biotic and abiotic pores (Duan et al., 2023). Among them, the function of vegetation roots is the main reason for the formation of biological pores. A survey of Chinese arboreal forests revealed that root biomass constitutes approximately 23 % of aboveground biomass (Ma, 1989). Coincidentally, research indicates that the biomass of black locust undergoes changes corresponding to forest age and density variations, with root biomass reaching up to 25 % of the total biomass under average conditions in artificial black locust forests in the semi-arid areas of the Loess Plateau (Cheng et al., 2013). During the vegetation restoration process, plant roots fully contact the soil and intersperse, entangle, divide and squeeze the soil, forming large soil pores, which increase soil porosity and improve soil permeability (Fischer et al., 2014). After the partial root degenerates and dies, the root pores formed by decay and decomposition increase soil porosity and promote the preferential flow of soil water (Beven and Germann, 1982; Luna et al., 2018). The proportion of these rotten root pores in forest soil is about 35 % (Zhang et al., 2017). With such a huge root system, the preferential flow channels formed by root development and rotten root channels promote a wider distribution and higher degree of differentiation of preferential flow in forest soil, which vitally influences the hydrological cycle and runoff generation mechanism of forest ecosystem (Lejeune et al., 2004; Liu et al., 2021).

As the forest age increases and the roots extend and grow, the total root biomass will increase. In addition, regional climate and soil conditions can influence root biomass (Li and Liu, 2014; Cui et al., 2020). The findings from a long-term monitoring test conducted on the Loess Plateau revealed that the average root biomass of 26-year-old black locusts is approximately 8.1 times that of 5-year-old black locusts in the 0–50 cm soil layer (Wang et al., 2007). The rich root system structure increases the possibility of forming large pores at the root-soil interface. However, when roots penetrate deep into the soil, they tend to occupy the existing large pores, causing the root hole structural characteristics to easily change with the planting years (Huang et al., 2019). In addition, the number of rotten root channels produced by the root function increases, especially the fine roots with short growth cycles and fast decay rates (Duan et al., 2023). The existence of rotten root channels significantly increases the density and connectivity of the macropore network, which is conducive to the occurrence of preferential flow (Wu et al., 2017; Nespoulous et al., 2019). Under rainfall conditions, deep and connected root pores structure is beneficial to deep infiltration of water and increases the water storage capacity of deep soil, thereby promoting water supply to the forest ecosystem (Pagenkemper et al., 2013). Research showed that compared with bare soil, the soil water storage capacity increased by rotten root channels can reach 37 %–260 % (Wu et al., 2021a). Forest root activity is a key factor influencing the development of preferential flow paths. A study identified that the abundant root in mixed forests within Karst areas could be a contributing factor to the higher development and variability of preferential flow compared to that in pure forests (Guan et al., 2023). Additionally, the generation of preferential flow paths in deeper layers has been attributed to root activities, as evidenced by Cheng et al. (2014) in their comparison of young and older forest plantations in the central portion of the Simian Mountains. Different root system traits also exert varying effects on the occurrence of preferential flow, several studies have indicated that fine roots play a significant role in promoting preferential flow (Luo et al., 2019b; Zhang et al., 2015). Current research primarily emphasizes the impact of fine roots on preferential flow, overlooking the

contributions of thick roots and decomposed roots. Changes in soil structure caused by forest root activity are key factors influencing the development of preferential flow paths. The development of the root brings about alterations in soil structure, which could potentially serve as a crucial factor in gauging the success of ecological restoration (Costantini et al., 2016; Bandyopadhyay and Maiti, 2019). Changes in soil structure caused by forest root activity are key factors influencing the development of preferential flow paths.

Artificial arbor forests are main body of the soil and water conservation forest system on the Loess Plateau, with the characteristic of developed root systems, rich biomass, and large storage capacity. Changes in macropores caused by root growth, decay, and death will affect the preferential flow and thus affect the deep soil moisture recharge, hydrological cycle and runoff generation mechanism of forest ecosystem. Comprehensive studies on the influence of root systems with different diameters on preferential flow are lacking, and the impact of the morphology of decayed roots in forest soil on preferential flow remains unexplored. The purpose of this study is to: (1) clarify the characteristic distribution of living roots and rotten roots of different forest ages, (2) analyze the influence of roots on preferential flow, and (3) explore the differences in preferential flow under distinct forest ages. These results will support further research on the relationship and rules between root functions and runoff generation mechanism of artificial forests, reveal the transformation relationship of water, prevent water and soil erosion, and provide a scientific basis for reasonably and effectively transforming and restoring vegetation and improving soil water storage.

## 2. Materials and methods

### 2.1. Study area and experiment design

This study was conducted in black locust plantations in the Zhifanggou watershed (109°13'03"–109°16'46"E, 36°46'28"–36°46'42"N) in the Loess Plateau in China from July 2022 to September 2022. The Zhifanggou watershed, located in the loess hilly gully area, is a typical catchment area with a slope of 0–65° and an altitude of 1010–1431 m (Zhao et al., 2016; Ding et al., 2021). Based on the monthly site data from 2005 to 2021, the average annual rainfall in this area is 481 mm, the average annual temperature is 9.8 °C, and the annual evaporation is 721 mm (<https://asa.cern.ac.cn/>). The soil type in the study area is loess soil, consisting of sandy soil (21 ± 6 %), silt soil (63 ± 3 %), and clay soil (16 ± 4 %) (Ye et al., 2018). According to the national standard LY2908-2017 classification standards for age classes and age groups of major tree species, the artificial black locust forest in the study area was divided into three groups: (1) young forest (≤15 years), (2) middle-aged forest (16–25 years), and (3) mature forest (31–40 years). The locations of forests with different stand ages were illustrated in Fig. 1. In each forest area, three 20 m × 20 m plots were established. Within each plot, three standard trees exhibiting similar growth patterns were randomly chosen. The fundamental information of these selected trees is presented in Table 1. The root collection test and dye-tracer infiltration test were conducted at a position 0.5 m away from the tree trunk. The soil properties of three different forest ages were determined using the cutting ring method (Table 2).

### 2.2. Root characterization determination

Nine trees of each forest age were used for root characteristic analysis. The characteristics of the root system were converted by the sectional method. The total sampling depth was 100 cm, divided into five layers with intervals of 20 cm. The sampling volume of root-soil samples in each soil layer is 50 cm × 20 cm × 20 cm (length × width × height). Root sampling was performed on the soil profile for the infiltration experiment. All root-soil mixture samples were elutriated in the laboratory to isolate roots. According to the root diameter of living

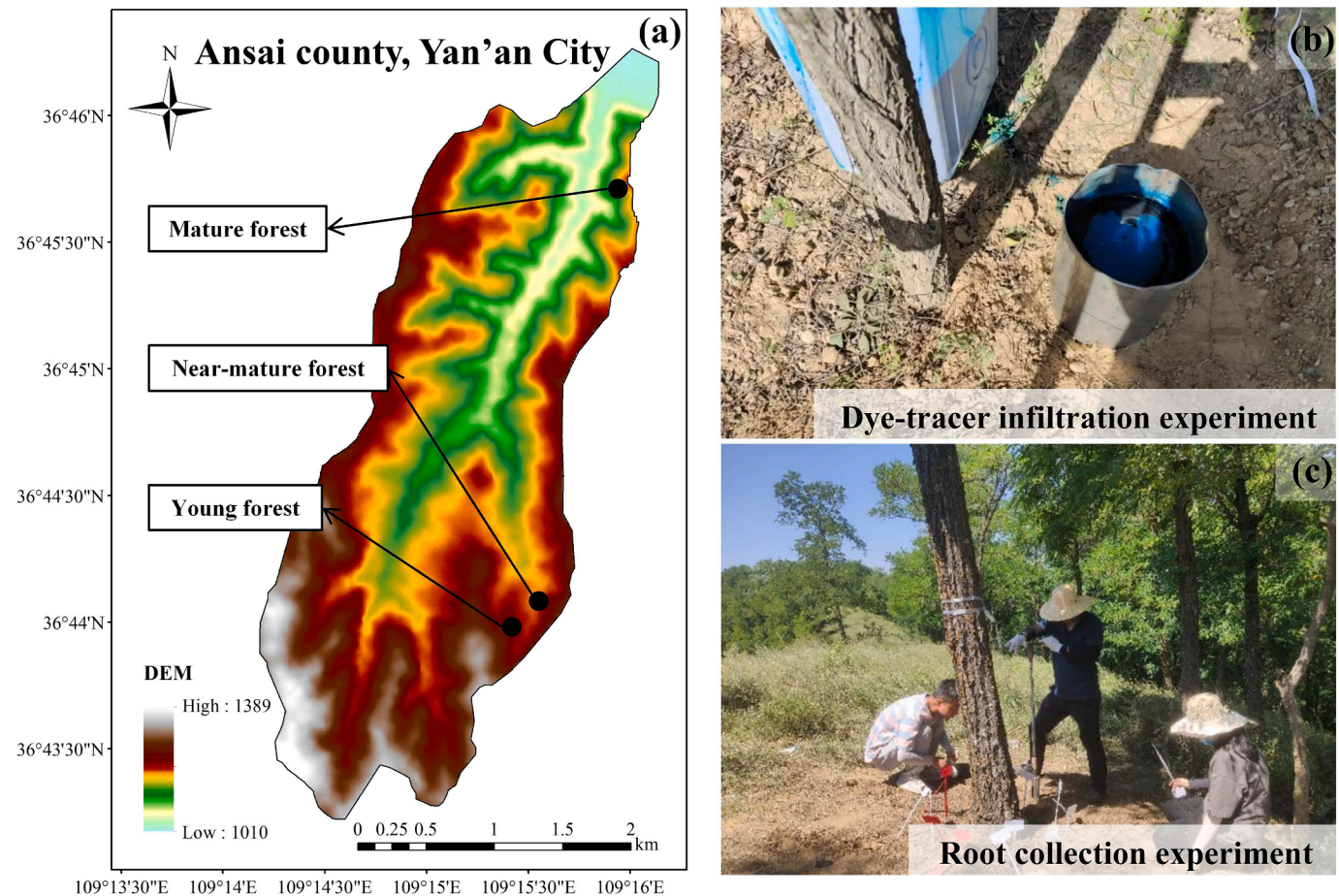


Fig. 1. Geographical location of the test area.

Table 1  
Basic characteristics of experimental sample plots.

| Type               | Stand age (a) | Height of tree (m) | Diameter at breast height (cm) | Crown diameter (m) | Slope (°) |
|--------------------|---------------|--------------------|--------------------------------|--------------------|-----------|
| Young forest       | 10–13         | 6.08 ± 0.63        | 11.03 ± 1.36                   | 1.93 ± 0.06        | 7         |
| Middle-aged forest | 19–22         | 9.29 ± 0.44        | 27.97 ± 1.14                   | 2.51 ± 0.17        | 5         |
| Mature forest      | 36–39         | 11.08 ± 0.90       | 32.28 ± 2.62                   | 4.09 ± 0.12        | 9         |

roots, the following classifications were carried out: <2 mm, 2–5 mm, 5–10 mm, 10–15 mm, 15–20 mm, 20–30 mm, 30–40 mm, 40–50 mm, 50–60 mm, 60–70 mm, 70–80 mm, and 80–90 mm. The root volume ( $V_{\text{roots}}$ ,  $\text{m}^3$ ) was determined through the drainage method. Prepare a container large enough to fully accommodate the roots to be tested. Fill the container with water, noting the initial water level. Carefully immerse the root system to be tested into the water, ensuring complete

submersion without the formation of air bubbles or floating on the water surface. Record the final water level after placing the roots in the water. Root length ( $L_{\text{roots}}$ , m) was measured using electronic length measuring tool. Specifically, the length of fine roots (<2 mm) was measured using a root scanner (LA-S). After the root volume measurement, all root samples were dried in an oven (65 °C) to constant weight ( $W_{\text{roots}}$ , kg).

$$RVR = \frac{V_{\text{root}}}{V_{\text{soil}}} \quad (1)$$

$$RLD = \frac{L_{\text{root}}}{V_{\text{soil}}} \quad (2)$$

$$RB = \frac{W_{\text{root}}}{V_{\text{soil}}} \quad (3)$$

where RVR is root volume ratio ( $\text{m}^3/\text{m}^3$ ), RLD is root volume ratio ( $\text{m}/\text{m}^3$ ), RB is root biomass ( $\text{kg}/\text{m}^3$ ).  $V_{\text{soil}}$  is the volume of the soil layer, whose value is  $2 \times 10^{-2} \text{ m}^3$ .

In addition to living roots, there were also many rotten roots in the soil. Based on the classification of the rotten roots of black locust forest

Table 2  
Soil properties of three different forest ages.

| Type               | Bulk density ( $\text{g}/\text{cm}^3$ ) | Total porosity (%) | Capillary porosity (%) | Initial gravimetric water content(%) | Capillary holding capacity (%) | Saturated water capacity (%) |
|--------------------|-----------------------------------------|--------------------|------------------------|--------------------------------------|--------------------------------|------------------------------|
| Young forest       | 1.2268 ± 0.0878a                        | 50.08 ± 1.29b      | 25.49 ± 1.29b          | 13.08 ± 0.29b                        | 23.87 ± 1.34b                  | 36.71 ± 3.13a                |
| Middle-aged forest | 1.1564 ± 0.0354ab                       | 53.71 ± 3.31a      | 29.36 ± 3.50a          | 12.91 ± 0.34b                        | 22.26 ± 0.88b                  | 32.25 ± 2.30b                |
| Mature forest      | 1.1403 ± 0.0661b                        | 56.36 ± 1.33a      | 29.67 ± 1.80a          | 13.47 ± 0.13a                        | 25.65 ± 1.09a                  | 34.70 ± 3.08ab               |



collected in field experiments, the rotten root types are roughly divided into the following three categories. Part of the rotten roots decay in the soil to form a loose and porous structure (Fig. 2-b(i)), and as the degree of decay deepens, they become humus that supplies the growth of plants. In another part of the rotten root system, the woody part gradually shrinks and degrades, but its root bark has good toughness, so a rotten root channel is formed between the woody part and the root bark (Fig. 2-b (ii, iii)).

For root systems that formed obvious decayed root channels, the diameter and length of the rotten root were measured. Rotten root average diameter (RARD, cm), rotten root length density (RRLD, m/m<sup>3</sup>) and rotten root channel volume (RRCV, m<sup>3</sup>/m<sup>3</sup>) were calculated accordingly (Guo, 2020), and the calculation formula is as follows:

$$RARD = \frac{\sum_{i=1}^n RRD_i}{n} \quad (4)$$

$$RRLD = \frac{\sum_{i=1}^n RRL_i}{V_{soil}} \quad (5)$$

$$RRCV = \frac{\sum_{i=1}^n (\pi \times RRD_i^2 / 4 \times RRL_i)}{V_{soil}} \quad (6)$$

where  $RRD_i$  and  $RRL_i$  are the diameter (m) and length (m) of the  $i$ -th rotten root, respectively.

### 2.3. Dye-tracer infiltration experiment

Nine dye-tracing experiments were carried out in sites representing young forest, middle-aged forest, and mature forest areas to visualize water flow paths. The experimental steps of dye-tracer infiltration experiment were carried out according to the procedures of Chen et al (2021). First, clean the litter and gravel on the soil surface, and then smooth the soil surface. Slowly smash a stainless steel ring into the soil 5 cm, with the height above the ground set at 25 cm, ensuring that the cylinder wall and the soil remain vertical. Then uniformly pour 4.0 g/L brilliant blue FCF dye solution 25 L into the ring frame, while ensuring that there is no overflow of the dye solution on the side walls and edges. Continue to monitor water level changes continuously throughout the trial to promptly identify any indications of lateral flow. Subsequently, record the total penetration time, and seal the ring to prevent evaporation. 24 h after the infiltration is completed, gently remove the ring. Excavate a soil profile of 100 cm × 100 cm with the center of the ring as the focal point, facing the direction of the tree trunk. The stained areas are all within the section. Calibrated steel strips were placed along the soil profile for reference, and the dyeing pattern of the section was recorded and photographed using a camera (OPPO find X3 model mobile phone). A total of nine dyeing experiments were conducted on black locust forest plots of different forest ages.

Use the unsupervised classification tool K-Means in ENVI 5.3 to identify the stained areas of the profile. To differentiate the dyed boundaries and ranges of different concentrations in the image, 20 categories were configured in the settings panel, and the initial classification image was generated through 15 iterations. The obtained preliminary classification results were visually interpreted. During the identification process, special attention should be given to exclude stained soil attached to the living root interface from the stained area. The determined stained areas are divided into two categories based on stain concentration. Active water zones mainly characterized by preferential flow exhibit dye concentrations greater than 2.0 g/L, while water buffer zones less than 2.0 g/L (Jiang et al., 2018; Chen et al., 2021). Set cyan to represent the buffer zone, blue to represent the active zone, and white to represent non-stained areas, such as soil and roots. This enhances the visual clarity of soil profile staining information for different forest ages in the image. ArcGIS10.3 software (ESRI Inc., Redlands, CA, USA) was used to calculate the areas of dye concentration, which was defined by shapefiles. The shapefile was divided into a grid

with 60 rectangular polygons of 160 cm<sup>2</sup> (80 cm wide and 2 cm high).

In addition, 20 × 120 cm dye-stained images of different forest ages were analyzed using ArcGIS 10.3 and decoupled from all dye-stained images to obtain parameters as priority flow indicators. The dye coverage (DC, %) of different forest ages calculation formula is as follows:

$$DC = \frac{DS}{DS + ND} \times 100\% \quad (7)$$

where DS and ND are the dyeing area (cm<sup>2</sup>) and the non-dyeing area (cm<sup>2</sup>), respectively.

The preferential flow fraction (PF-fr, %) is the total penetration fraction through the preferential flow path, and the calculation formula is as follows:

$$PF-fr = 100 \times \left( 1 - \frac{UniFr \times 20}{ToStAr} \right) \quad (8)$$

where UniFr means uniform infiltration depth (cm), which the depth where DC drops below 80 %. TotStAr means total stained area (cm<sup>2</sup>).

Length index of preferential flow (LI) is the sum of absolute differences between DC values with depth in soil profile (Bargues et al., 2014), and the calculation formula is as follows:

$$LI = \sum_{i=1}^{60} |DC_{i+1} - DC_i| \quad (9)$$

where  $DC_i$  is the dyeing area ratios corresponding to soil profile layer  $i$ .

Peak index of preferential flow (PI) refers to the number of intersections between the average DC vertical line and the actual DC profile.

### 2.4. Statistical analysis

One-way analysis of variance (ANOVA) was used to evaluate the differences in the root characterization between the various soil layers and different forest types (Lix et al., 1996). This study also determined the effect of tree stand ages on the soil properties and preferential flow indices by performing ANOVA. The correlation between root characterizations and preferential flow indices and root characterizations of various RD and preferential flow indices were evaluated using Pearson correlation analysis.

## 3. Results

### 3.1. Root characteristics

#### 3.1.1. Living root characteristics

A variance analysis was conducted on the root characteristics at different vertical depths of each forest age and found that the variation in root characteristics of young, middle-aged, and mature forests showed a trend of initial rise followed by a decline with the increase of soil depth (Fig. 3). The average root biomass density in soil increased significantly with the growth of forest age, which were 1.1, 3.1, and 6.5 kg/m<sup>3</sup>, respectively (Fig. 3a). The average root length density of young forests (46.2 m/m<sup>3</sup>) was significantly lower than that of middle-aged forests (94.2 m/m<sup>3</sup>) and mature forests (86.2 m/m<sup>3</sup>) ( $P < 0.05$ ). The distribution of root length density in different soil depths of each forest age was consistent with root weight density (Fig. 3b). The average root volume ratios of different forest ages in forest soil were  $7.4 \times 10^{-3}$  m<sup>3</sup>/m<sup>3</sup>,  $16.1 \times 10^{-3}$  m<sup>3</sup>/m<sup>3</sup>, and  $27.9 \times 10^{-3}$  m<sup>3</sup>/m<sup>3</sup>, respectively, which all increased significantly with the growth of forest age. The root volume ratio of each forest age was centered in the soil depth of 20–40 cm, significantly higher than other soil layers ( $P < 0.05$ ) (Fig. 3c).

The distribution of root characteristic parameters of each root diameter in different soil layers was also quite different, which may be



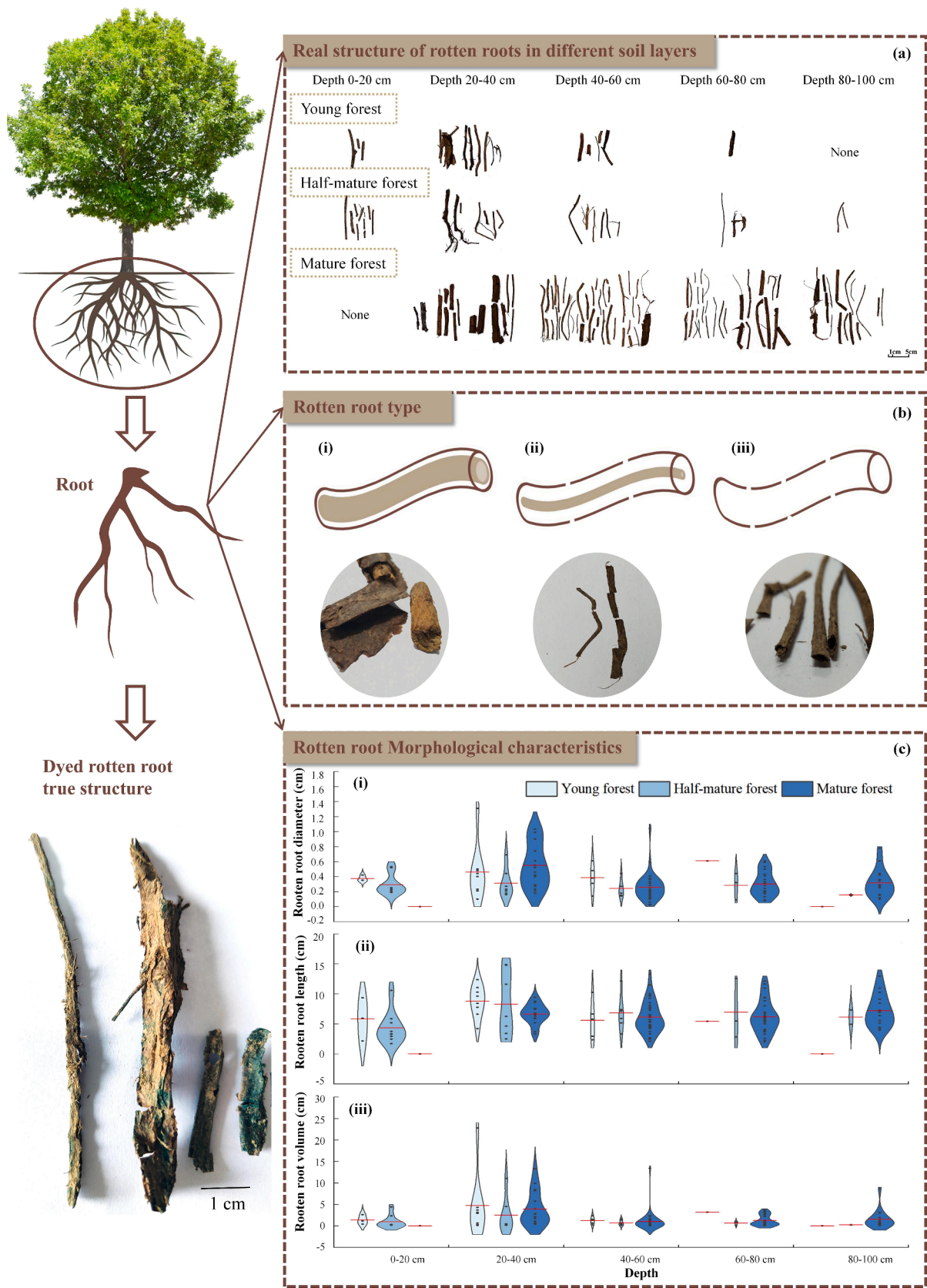
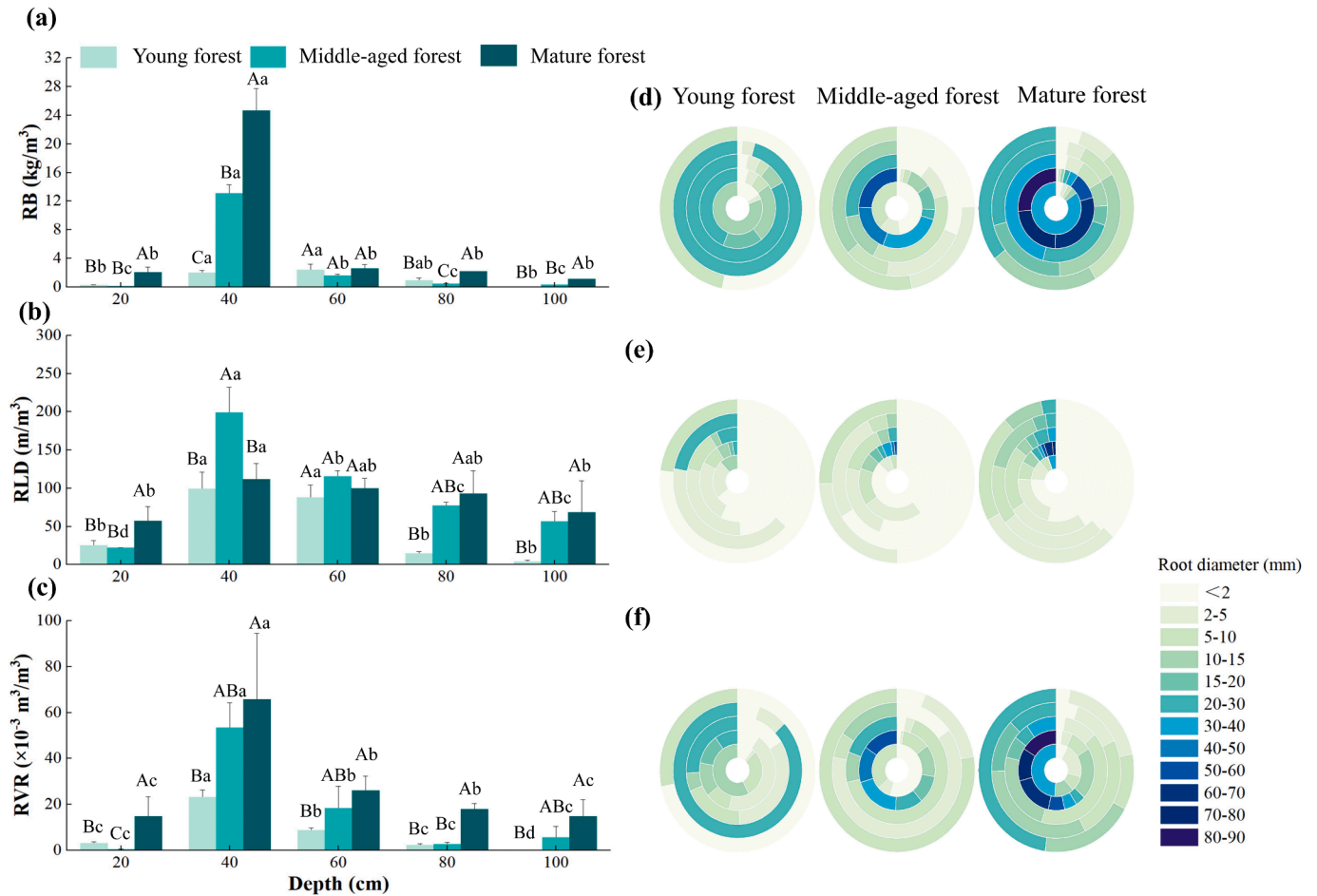


Fig. 2. Actual structure and morphology of rotten roots in black locust forest soil.



**Fig. 3.** Root characteristics and proportions of each diameter class in different soil depths of each stand ages. Among them, from the inside to the outside in the ring diagram are the proportions of root characteristics of each diameter class at depths of 0–20 cm, 20–40 cm, 40–60 cm, 60–80 cm, and 80–100 cm. Different uppercase letters indicate a significant difference among different forest types ( $P < 0.05$ ) and different lowercase letters indicate a significant difference among different soil layers ( $P < 0.05$ ).

caused by the differences in root diameters in each soil layer (Fig. 3d, e, f). The maximum root diameter in young forests could reach 20–30 mm, mainly distributed in the 20–80 cm soil layer. The root diameter in middle-aged forests can reach up to 50–60 mm, concentrated in the soil depth of 20–40 cm. In mature forests, the maximum diameter of the root system was also mainly distributed in the 20–40 cm soil layer, which could reach 80–90 mm.

In young forests, the dominant root diameter shifts with soil depth (Fig. 3d), 10–15 mm roots dominate in the 0–20 cm layer (80.7 %), 20–30 mm roots prevail in the 40–80 cm layer (43.8–95.6 %), and < 2 mm roots dominate in the 80–100 cm layer (53.2 %). In middle-aged forests, the dominant root diameters with the largest proportion of root biomass at different soil depths are < 2 mm (48.0 %, 0–20 cm), 30–40 mm (26.5 %, 20–40 cm), 5–10 mm (36.7 %, 40–60 cm), 10–15 mm (36.4 %, 60–80 cm), and 5–10 mm (53.1 %, 80–100 cm). In mature forests, the dominant root diameters with the largest proportion of root biomass at different depths are 30–40 mm (85.1 %, 0–20 cm), 60–70 mm (29.9 %, 20–40 cm), 30–40 mm (45.8 %, 40–60 cm), 10–15 mm (32.5 %, 60–80 cm), and 20–30 mm (35.5 %, 80–100 cm).

The root diameter with the highest proportion of root length density in each soil layer of different forest ages is < 2 mm, accounting for 36.6–77.1 % (young forest), 39.3–86.9 % (middle-aged forest), and 36.9–76.1 % (mature forest), respectively (Fig. 3e).

In young forests, the primary root diameters with the highest root volume ratio at various soil depths are: 10–15 mm (57.4 %, 0–20 cm), 2–5 mm (33.3–44.2 %, 20–60 cm), 20–30 mm (87.1 %, 60–80 cm), and

< 2 mm (71.8 %, 80–100 cm). In middle-aged forests, key root diameters with the highest root volume ratio at different soil depths are: < 2 mm (54.5 %, 0–20 cm), 30–40 mm (19.6 %, 20–40 cm), 5–10 mm (65.8 %, 40–60 cm), 2–5 mm (63.2 %, 60–80 cm), and 5–10 mm (77.7 %, 80–100 cm). In mature forests, primary root diameters with the highest root volume ratio at various soil depths are: 30–40 mm (49.2 %, 0–20 cm), 10–15 mm (20.7 %, 20–40 cm), 5–10 mm (41.1 %, 40–60 cm), 10–15 mm (32.2 %, 60–80 cm), and 20–30 mm (47.7 %, 80–100 cm) (Fig. 3f).

### 3.1.2. Rotten root characteristics

As the forest age increases, the number of rotten roots in the soil of black locust forest increases. As shown in Fig. 2a, most of the rotten roots in young and middle-aged forests were concentrated in the 0–60 cm soil layer. While in mature forests, there were no rotten roots at a depth of 0–20 cm, and rotten roots were concentrated in the 20–100 cm soil layer. Analyzing decayed root morphology (Fig. 2c), young forests show more favorable average length and channel volume in the topsoil (0–40 cm) than middle-aged and mature forests. No significant differences in rotten root morphology were observed at a depth of 40–80 cm among different forest ages. In deep soil (80–100 cm), mature forests exhibit superior characteristic decayed root morphology compared to middle-aged forests.

The average diameter of rotten roots at 20–40 cm in black locust forests of all ages was the largest, which were 0.46 cm (young forest), 0.31 cm (middle-aged forest), and 0.55 cm (mature forest), respectively. However, the length density of rotten roots in black locust forests of

different ages showed a trend of initial increase followed by a decrease with the deeper of soil depth. The length density of rotten roots in young and middle-aged forests was the largest at a soil depth of 20–40 cm, respectively. It was  $35.2 \text{ m/m}^3$  (young forest) and  $29.0 \text{ m/m}^3$  (middle-aged forest). In mature forests, the rotten root length density was the highest at 40–60 cm soil depth, which was  $116.6 \text{ m/m}^3$ . The changing trend of rotten root channel volume ratio with soil layer in young and middle-aged forests was similar to that of rotten root length density. Both of them had the maximum rotten root channel volume ratio at 20–40 cm soil depth, which was  $1.89 \times 10^{-3} \text{ m}^3/\text{m}^3$  (young forest) and  $0.87 \times 10^{-3} \text{ m}^3/\text{m}^3$  (middle-aged forest), respectively. The volume ratio of rotten root channels in mature forests decreased with the increase of soil depth and is the largest at 20–40 cm, with a value of  $3.15 \times 10^{-3} \text{ m}^3/\text{m}^3$  (Table 3).

### 3.2. Preferential flow characteristics and differences

The flow pattern was visually represented on the vertical soil profile through the dye tracing method (Fig. 4). The dyeing zone was evenly distributed in the depth of 0–20 cm in young and middle-aged forests, while the stained areas in the 0–30 cm soil layer in mature forests were evenly distributed, and the stained patches in the remaining soil layers were more scattered. Through variance analysis on the preferential flow characteristic parameters (Table 4), it was found that the total infiltration time of young forests (338.0 min) was obviously greater than that of middle-aged forests (271.3 min) and mature forests (315.3 min). The TotStAr on the vertical profile of the soil decreased significantly with the increase of forest age, which were  $935.3 \text{ cm}^2$  (young forest),  $1369.9 \text{ cm}^2$  (middle-aged forest), and  $1718.1 \text{ cm}^2$  (mature forest), respectively. The changing trend of active water area with forest age was consistent with the total dyeing area, but the area of water buffer zone had no significant statistical significance ( $P > 0.05$ ). In addition, the maximum infiltration depth and width showed a downward trend with the increase of forest age.

All parameters that quantify the degree of preferential flow found a relatively consistent trend (Fig. 5). Among them, the DC of young forests (62.7 %) and middle-aged forests (63.0 %) were not significantly different, but they were both substantially lower than that of mature forests (78.2 %,  $P < 0.05$ ). The PF-fr of young forests was obviously higher than that of middle-aged forests ( $P < 0.01$ ), which in turn was substantially greater than that of mature forests ( $P < 0.05$ ). The UniFr of young forests was considerably lower than that of middle-aged forests ( $P < 0.01$ ), and the latter was substantially lower than that of mature forests ( $P < 0.01$ ). The LI of mature forests was obviously higher than that of young and middle-aged forests ( $P < 0.05$ ), indicating that the infiltration paths of young and middle-aged forests were more irregular, while they were more uniform in mature forests.

### 3.3. Effect of roots on preferential flow

An analysis of the correlation between the root characteristics of living roots and rotten roots and the preferential flow characteristic index found that preferential flow characteristic index was related to RARD, RRLD, and RVR (Fig. 6a), which existed a significant correlation among them ( $P < 0.05$ ). The average diameter of rot roots was substantially negatively correlated with UniFr ( $P < 0.05$ ) and significantly positive associated with PF-fr ( $P < 0.05$ ); the root length density of rot roots was extremely significantly positive associated with DC, TotStAr, and UniFr ( $P < 0.001$ ), showing a notable adverse correlation with PF-fr ( $P < 0.01$ ) and LI ( $P < 0.05$ ); rot root volume ratio showed a strong positive relationship with DC ( $P < 0.001$ ), TotStAr ( $P < 0.05$ ), and UniFr ( $P < 0.05$ ), and a substantial negative association with LI ( $P < 0.01$ ).

Among the preferential flow characteristic indicators, DC and RVR reached a substantial positive relationship ( $P < 0.05$ ). The correlation between root structure characteristics of different diameters and preferential flow characteristic indicators was further analyzed. The results showed that the relationship between the preferential flow characteristic indices and the RB of different diameter classes was mainly concentrated in the root diameter of 30–90 mm (Fig. 6b). Among them, the RB of 5–10 mm diameter was closely related to PF-fr, showing a strong inverse relationship ( $P < 0.01$ ). The RB of 30–40 mm diameter was vital positive associated with DC ( $P < 0.01$ ), TotStAr ( $P < 0.05$ ) and UniFr ( $P < 0.05$ ). The RB of 50–90 mm diameter exhibited a significant positive correlation with DC ( $P < 0.05$ ). The relationship between the preferential flow characteristic indices and the RLD of different diameters was similar to the correlation with RB (Fig. 6c). The RLD of roots in the 5–10 mm diameter class exhibited notable adverse correlation with PF-fr ( $P < 0.05$ ); the RLD of 30–40 mm diameter shows a strong positively correlated with DC ( $P < 0.01$ ), TotStAr ( $P < 0.01$ ), and UniFr ( $P < 0.05$ ); the RLD of 50–90 mm diameter had a significant positive relationship with DC ( $P < 0.05$ ). The link between the preferential flow characteristic indexes and the RVR of different diameters was shown in Fig. 6d, in which the RVR of 2–5 mm diameter had a substantial positive correlation with DC ( $P < 0.05$ ); the correlation between RVR of 5–10 mm diameter and PF-fr was pronounced negative ( $P < 0.05$ ); the RVR of the 30–40 mm diameter root system exhibited a substantial negative relationship with DC ( $P < 0.001$ ), TotStAr ( $P < 0.05$ ), and a strong positive relationship with UniFr ( $P < 0.05$ ); the RVR of 50–90 mm diameter was considerably positively correlated with DC ( $P < 0.05$ ).

## 4. Discussion

### 4.1. Vertical distribution root characteristics of different forest ages

The vertical distribution of roots in the soil varies greatly with the degree of growth, development, decline, and aging. This study found that as the forest age increased, the root characteristic parameters (RB,

**Table 3**  
Characteristics of black locust roots of different ages in soil profile.

| Depth<br>(cm) | Average rotten root diameter<br>(cm) |                    |                    | Rotten root length density<br>( $\text{m/m}^3$ ) |                   |                    | Rotten root volume ratio<br>( $\times 10^{-3} \text{ m}^3/\text{m}^3$ ) |                    |                    |
|---------------|--------------------------------------|--------------------|--------------------|--------------------------------------------------|-------------------|--------------------|-------------------------------------------------------------------------|--------------------|--------------------|
|               | S1                                   | S2                 | S3                 | S1                                               | S2                | S3                 | S1                                                                      | S2                 | S3                 |
| 0–20          | 0.3743 ±<br>0.0414                   | 0.2924 ±<br>0.1334 | —                  | 8.730 ±<br>1.711                                 | 19.545 ±<br>1.335 | —                  | 0.2093 ±<br>0.0448                                                      | 0.4270 ±<br>0.0737 | —                  |
| 20–40         | 0.4584 ±<br>0.3735                   | 0.3108 ±<br>0.1900 | 0.5498 ±<br>0.2986 | 35.160 ±<br>2.439                                | 29.010 ±<br>2.716 | 53.145 ±<br>0.829  | 1.8922 ±<br>0.2831                                                      | 0.8658 ±<br>0.1438 | 3.1497 ±<br>0.2000 |
| 40–60         | 0.3839 ±<br>0.1764                   | 0.2416 ±<br>0.1115 | 0.2580 ±<br>0.1765 | 13.980 ±<br>1.091                                | 20.535 ±<br>1.379 | 116.600 ±<br>1.293 | 0.3056 ±<br>0.0251                                                      | 0.2157 ±<br>0.0174 | 1.9494 ±<br>0.1094 |
| 60–80         | 0.6112                               | 0.2839 ±<br>0.1792 | 0.2997 ±<br>0.1617 | 2.715                                            | 10.425 ±<br>1.290 | 86.460 ±<br>1.358  | 0.1593                                                                  | 0.0958 ±<br>0.0116 | 1.7502 ±<br>0.0625 |
| 80–100        | —                                    | 0.1537 ±<br>0.0095 | 0.3139 ±<br>0.1900 | —                                                | 6.135 ±<br>1.102  | 53.970 ±<br>1.341  | —                                                                       | 0.0224 ±<br>0.0040 | 1.1571 ±<br>0.1112 |



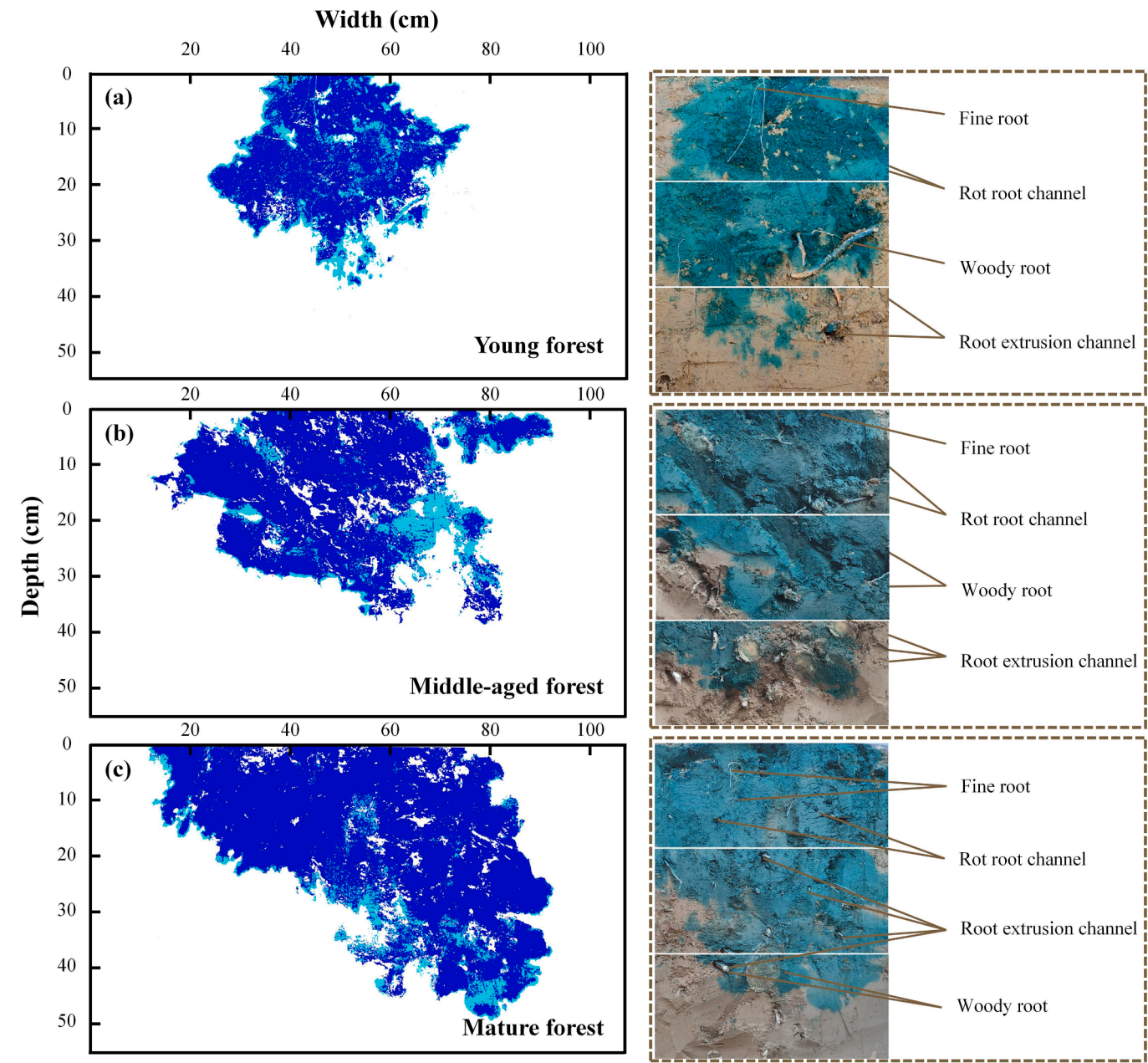


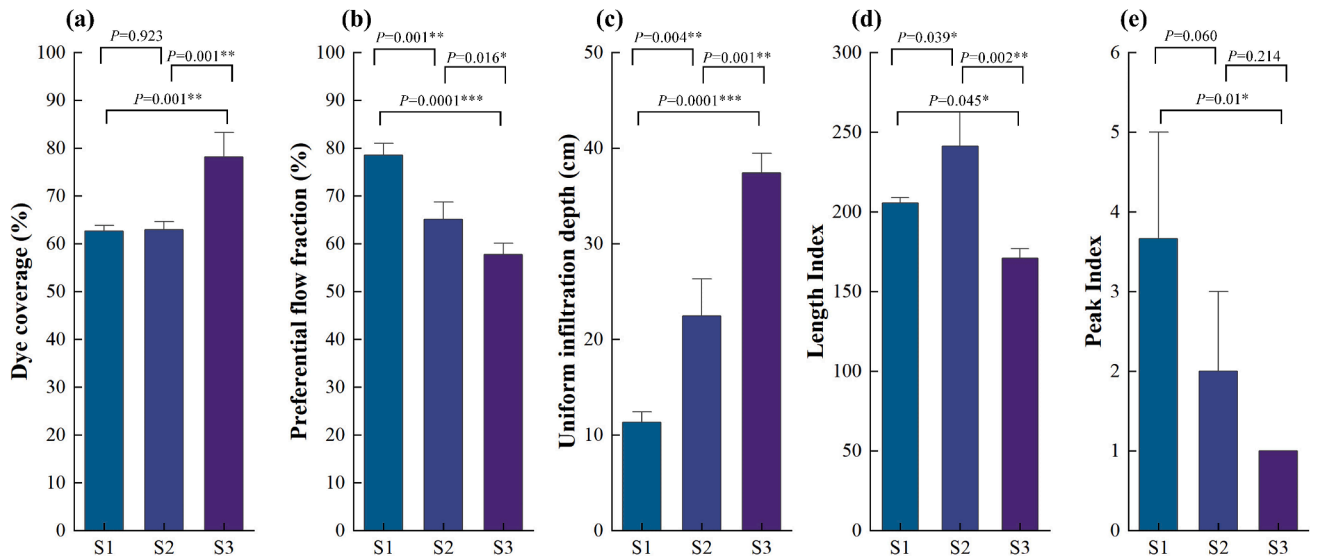
Fig. 4. Dye dye infiltration patterns and true soil structure in the vertical soil sections of young forests (a), middle-aged forests (b) and mature forests (c).

**Table 4**  
Dye staining characteristics in vertical dye soil profiles of black locust forests of different stand ages.

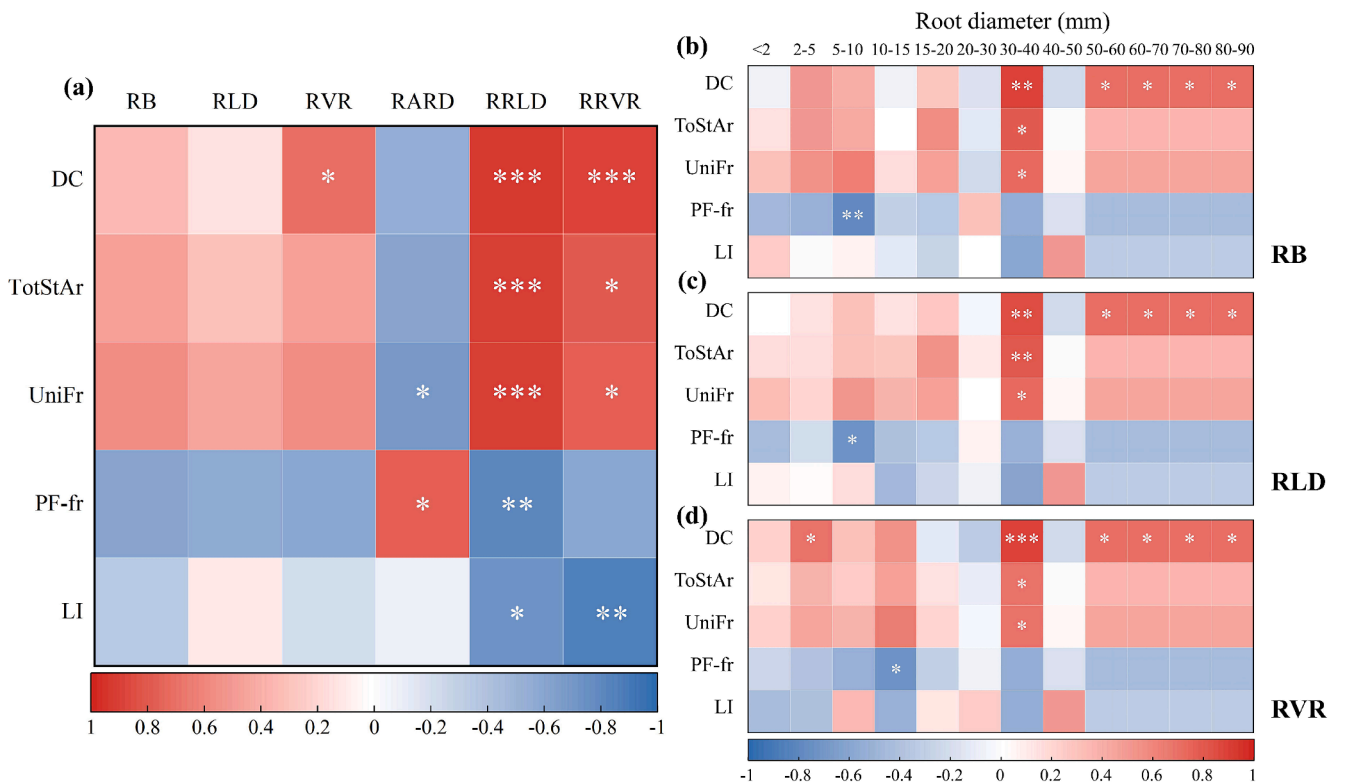
| Type | Dye solution infiltration rate<br>(cm <sup>3</sup> /min) | Dye solution volume<br>(L) | Maximum dye-stained depth<br>(cm) | Maximum dye-stained width<br>(cm) | Total stained area<br>(cm <sup>2</sup> ) | Active water zones<br>(cm <sup>2</sup> ) | Water buffer zones<br>(cm <sup>2</sup> ) |
|------|----------------------------------------------------------|----------------------------|-----------------------------------|-----------------------------------|------------------------------------------|------------------------------------------|------------------------------------------|
| S1   | 73.97b                                                   | 25                         | 37.7 ± 2.8b                       | 52.2 ± 1.6b                       | 935.3 ± 103.9c                           | 677.1 ± 119.7c                           | 258.8 ± 16.3a                            |
| S2   | 92.14a                                                   | 25                         | 39.0 ± 4.1b                       | 79.5 ± 11.9a                      | 1369.9 ± 146.3b                          | 1037.5 ± 159.4b                          | 332.0 ± 92.2a                            |
| S3   | 79.28b                                                   | 25                         | 49.3 ± 4.9a                       | 79.4 ± 5.3a                       | 1718.1 ± 179.4a                          | 1448.3 ± 220.8a                          | 269.9 ± 42.3a                            |

RLD, and RVR) also increased. Compared to middle-aged and mature forests, young forests with less understory litter, low soil microbial diversity, and poor soil nutrients, (Xu et al., 2020, Zhang et al., 2022), resulting in slow root development and growth, and most of them are concentrated in the surface soil (Chang et al., 2017). As the artificial

black locust forest land restoration process advances, substantial changes occur in canopy density, understory vegetation composition and diversity, as well as litter and soil nutrients on the Loess Plateau (Xu et al., 2020). There is an increase in species diversity, thriving micro-organisms, and enhanced nutrient levels in the soil (Kou et al., 2016, Liu



**Fig. 5.** Parameter characteristics of (a) dye coverage, (b) preferential flow fraction, (c) uniform penetration depth, (d) length index and (e) peak index of preferential flow in young forests, middle-aged forests and mature forests. S1: young forest; S2: middle-aged forest; S3: mature forest. Different lowercase letters indicate a significant difference at  $P < 0.05$ .



**Fig. 6.** Correlation coefficients among preferential flow characteristic parameters and (a) root structure characteristics of living roots and rotten roots, (b) root weight density of different diameter classes, (c) root length density of different diameter classes and (d) root volume ratio of different diameter classes. The red color represents positive correlations, and the blue color represents the negative correlations. The color intensity of the blocks is proportional to the correlation coefficients. In the right side of the correlogram, the legend colour shows the correlation coefficients and the corresponding colours. DC: dye coverage (%); ToStAr: total stained area (cm<sup>2</sup>); UniFr: uniform penetration depth (cm); PF-fr: preferential flow fraction(%); LI: length index; RB: root biomass (kg/m<sup>3</sup>); RLD: root length density (m/m<sup>3</sup>); RVR: root volume ratio (m<sup>3</sup>/m<sup>3</sup>); RARD: rotten root average diameter (cm); RRLD: rotten root channel length density (m/m<sup>3</sup>); RRRV: rotten root channel volume ratio (m<sup>3</sup>/m<sup>3</sup>). \* $P < 0.05$ , \*\* $P < 0.01$ , \*\*\* $P < 0.001$ . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

et al., 2018a). All of these not only supply the above-ground growth and development of trees, but also provide nutrients to the roots and increases their biomass (Xu et al., 2010). Moreover, the growth and

development of above-ground parts intensify plant transpiration. In order to supplement the water needed for tree growth in the soil of arid areas, the root system will continue to extend into the deep soil to absorb

water (Jia and Shao, 2014), causing the root biomass in the deep soil to increase.

During the forestland restoration process, as the root system continues to extend, root system turnover, that is, the decline and death of root system will continue to intensify (Fig. 2). In this study, the average volume of rotten root channels in mature black locust forests can reach  $1.60 \times 10^{-3} \text{ cm}^3/\text{cm}^3$ , which was 3.1 times that in young forests and 4.9 times that in middle-aged forests. Studies have shown that the rate of root decomposition mainly depends on root characteristics and the decomposition environment of the root system (Freschet et al., 2013, Zhang and Wang, 2015). This study found that the average diameter width of rot roots varied from 0.15 to 0.61 cm. The size of the root system diameter determines the speed of root decomposition (Tang et al., 2015, Liu et al., 2023). The higher is the content of root lignin, cellulose, hemicellulose and others, the slower is the root decomposition rate (Berg et al., 1993, Yang et al., 2004). In addition, this study found that the rot roots of young forests were mostly concentrated in the 0–60 cm soil layer, the rot roots of middle-aged forests mainly appeared at the soil depth of 0–80 cm, and the mature forests had no rot roots at the soil depth of 0–20 cm. There were rotten roots in mature forests, concentrated in the 20–100 cm soil layer. This may be because young and middle-aged forests are in the early stages of vegetation production, and root systems are mostly concentrated in the middle and upper layers of the soil. During the development process of roots, some roots decline or even die due to competition for nutrients and water. With the development of vegetation, the diversity of above-ground vegetation increases, litter is abundant and mostly concentrates on the soil surface, and the surface soil has better aeration with rich soil nutrients (Bradford et al., 2013, Sokol and Bradford, 2019, Prescott and Vesterdal, 2021). The enhancement of soil structure with increasing forest age (Table 2), thereby facilitating the supply of water and nutrients for plant root growth. The reduced incidence of root decline and death in the soil surface in mature forests, compared to young and middle-aged forests, can be attributed to this factor.

#### 4.2. Factors contributing to the preferential flow

This study used dye-tracer method to visualize the pattern and degree of preferential flow in soil of different forest ages. The study found that as the forest age increased, the staining depth and total area continued to increase (Table 4). Zhao et al. (2023) found that with the artificial eucalyptus forest age increased, the preferential flow continued to increase. Cheng et al. (2014) also found that the preferential flow of soils of different forest ages had great differences in the vertical direction, and preferential flow depth increases with the forest age. This may be attributed to the soil matrix of different forest ages and root system differences (Cheng et al., 2014, Luo et al., 2019b, Zhao et al., 2023). As the forest matures, there is a notable decrease in soil bulk density accompanied by a significant increase in porosity (Table 2). Concurrently, indicators such as initial soil moisture content and capillary water holding capacity experience an augmentation, contributing to an overall improvement in soil quality. These enhancements in soil structure exert a positive influence on soil infiltration processes, consequently impacting preferential flow. The PI and LI values of mature forests were lower than those of young and middle-aged forests (Fig. 5), indicating that preferential flow pattern in mature forests was relatively uniform. This may be ascribed to the continuous development and extension of the root system during the vegetation restoration process, and the root channels are abundant and relatively evenly distributed, which promotes preferential inflow and infiltration in soil (Kan et al., 2020).

The root of forest is an important part of the soil layer, whose structural shape affects preferential flow process. The root channels of plantations with different restoration years may be a critical factor leading to the differences in preferential flow (Jiang et al., 2018). Roots can wrap and rearrange soil particles and release a large amount of

secretions, causing physical, chemical, and biological changes in soil properties, thereby affecting soil permeability (Kalhor et al., 2019, Shi et al., 2021). Moreover, root channels formed by root rot after plant death can significantly increase the soil porosity, thereby increasing the soil infiltration performance (Huang et al., 2016, Wu et al., 2016, Liu et al., 2020). Studies have shown that root structure can promote the occurrence of preferential flow and transport solutes to deep soil layers (Wu et al., 2021b, Luo et al., 2019c). This study found that RVR had a significant positive impact on staining area (Fig. 6a), which may be caused by stem flow infiltration along the roots (Guo et al., 2020).

Further analysis of the effects of root characteristics of different root diameters on preferential flow found that, the RVR of 2–5 mm had a significant positive effect on the DC, the roots of 30–40 mm and > 50 mm significantly promote preferential flow, while roots of 5–15 mm may have a potential inhibitory effect on preferential flow (Fig. 6b, c, d). The influence of the distribution characteristics of various root diameters on soil macropores may lead to differences in preferential flow (Zhang et al., 2015), in which fine roots will enhance preferential flow (Luo et al., 2019c). This may be owing to the root volume of fine roots shrinking under drought conditions (Nobel and Cui, 1992), and fine root diameter can even be reduced by up to 60 % (Huck et al., 1970), thereby forming channels at the root-soil interface and promoting preferential flow. The thicker roots serves as a supporting root system. Under prevailing wind conditions, the wind drives the trunk, causing the roots to move and displace the soil around the channel wall (Ghestem et al., 2011), and water flows along the pores between the thick roots and the soil (Guo et al., 2020), thereby promoting the occurrence of priority flow. The 5–15 mm root increases anchoring effect between root and soil due to the growth and extension of the fine roots around it, reducing the gap between the soil and root system caused by movement. Meanwhile, the root during the growth period continues to increase in diameter and will also increase the squeezing effect of roots in soil, thus inhibiting preferential flow.

Rotting root channels may be a key factor influencing the preferential inflow depth of woodland soils. This study found that the average diameter of rot roots, root length density, and root volume density all significantly promoted preferential flow (Fig. 6a). The pores left after the death and decay of plant roots will increase soil porosity and form important water-conducting channels in the soil, thereby accelerating soil water movement (Kiepe, 1995, Chisci et al., 2001). In arid environments, root channels can increase the flux of water infiltrating into deeper soil layers. As the length of root channels increases, the influence of root channels on the downward movement of water becomes more obvious (Devitt and Smith, 2002). In addition, root turnover becomes frequent as the forest age increases (Ghestem et al., 2011), forming more rotten root channels in the soil (Fig. 2a). This study showed that in mature forests, the volume of rotten root channels in the soil accounted for about 0.31 %, which was 1.66 times that in young forests and 3.63 times that in middle-aged forests with the soil depth of 20–40 cm (Table 2). Therefore, the preferential flow of mature forests can be transported to deeper soil. Moreover, the root pore channel shapes formed by different degrees of root rot are various, and the diameter of the effective channel of rotten roots is significantly different (Fig. 2b), which affecting the preferential flow (Wu et al., 2021a). As the forest age increases, the degree of root decayed and dead will intensify, forming larger pores and inducing preferential flow. A large amount of nutrients will also be released during root decomposition (Hobbie, 2015), and incompletely decomposed roots will also retain solutes in the preferential flow (Vintner et al., 1999). As a result, the incompletely decomposed root system can increase soil fertility, provide nutrients and substrate for microorganisms in the surrounding soil, and increase the activity and quantity of soil microorganisms near rotten roots, thus further improving the surrounding soil structure and promoting water infiltration (Wu et al., 2014, Wu et al., 2017). However, the specific effects of different forms of rotten root channels on soil moisture transport are still unclear, and further analysis and research are needed in the future.



### 4.3. Impact and prospects on circulation mechanism of artificial forestry

The Loess Plateau has criss-cross ravines, thick soil, and poor infiltration capacity, and most of the rainfall occurs in summer, with short duration and strong rainfall. As a result, the flooding process in this basin is characterized by peak height, small volume, short duration, and steep rise in flood peaks, steep drop (Wang and Jiao, 1996). Moreover, owing to poor underlying surface conditions, surface runoff cannot be effectively retained. Affected by terrain, soil, climate, and other aspects, the serious water and soil erosion problem in the Loess Plateau has always attracted much attention. Vegetation can regulate the watershed runoff mechanism by changing vegetation coverage, litter thickness, and root density (Farley et al., 2005). In the young black locust forests, the ecosystem is still a single artificially planted vegetation system, with a single vegetation structure, low vegetation coverage, and weak rainfall retention effect. There is too little litter on the underlying surface to effectively intercept water, and the root system develops slowly (Fig. 3). The soil structure and nutrients are still in the early stages of recovery. At the same time, there are few effective rotten root channels (Figs. 2, 4), and water flows preferentially. The degree of infiltration recharge is low (Fig. 4), so the soil near the surface is more erodible. With vegetation restoration, the simple artificially planted vegetation system has developed into a complex artificially planted natural ecosystem. The vegetation coverage has increased, effectively retaining rainfall and replenishing it in the soil. Meanwhile, abundant litter provides a large amount of humus for the surface soil and promotes microbial activity. As the root system grows, extends, and turns over, it not only improves the soil structure but also improves soil nutrients, gradually reducing the erodibility of the near-surface soil, which can effectively control soil erosion (Zhu et al., 2010; Wang et al., 2018). In addition, compared with surface soil, the physical properties of deep soil improve relatively slowly during the vegetation restoration period (Li and Shao, 2006). Therefore, in the early stages of artificial vegetation restoration, more attention should be paid to the interception effect of a single vegetation canopy and litter to improve the water storage function of the soil, thereby effectively preventing surface water and soil erosion. At the same time, relevant research on the impact of roots on deep soil structure and nutrients during the root turnover process should be carried out to further clarify the influence of root effects on soil infiltration and preferential flow, thereby providing effective and scientific management measures for water replenishment of deep soil in artificial black locust forests.

## 5. Conclusions

This study evaluated the effects of different stand ages on preferential flow, including living root characteristics, rotten root channel characteristics, and preferential flow patterns. The staining area, staining rate, and uniform penetration depth increased with planting years increase. The research results showed that: (1) Root systems of each forest age were mainly distributed in the soil depth of 0–60 cm, the root biomass, root length density, and root volume ratio of young forests were significantly lower than those of middle-aged and mature forests ( $P < 0.05$ ), and the root characteristics of different forest ages and diameter classes were also significantly different. (2) Rotten roots in young and middle-aged forests were mainly distributed at a soil depth of 0–60 cm, while rot roots in mature forests were distributed at a depth of 20–100 cm. The volume density of rotten root channels in mature black locust forests was  $1.60 \times 10^{-3} \text{ m}^3/\text{m}^3$ , which was 3.1 times that in young forests and 4.9 times that in middle-aged forests. (3) The preferential flow (DC, TotStAr, and UniFr) of artificial black locust forests increased with the increase of forest age. Roots of 2–5 mm, 30–40 mm, and > 50 mm had a significant promoting effect on preferential flow, while roots of 5–10 mm inhibited preferential flow. The average diameter of rotten roots, root length density, and root volume density all significantly promoted preferential flow. These data can help to understand the water

recharge and infiltration conditions, such as preferential flow under the action of roots during the vegetation restoration process on the Loess Plateau, and provide a reference for research on the surface runoff generation mechanism of artificial forest ecosystems.

## CRediT authorship contribution statement

**Lu Cai:** Data curation, Formal analysis, Visualization, Writing – original draft. **Fei Wang:** Conceptualization, Funding acquisition, Writing – review & editing. **Yuanyuan Lin:** Methodology, Writing – original draft. **Qi Long:** Writing – original draft, Writing – review & editing. **Yulong Zhao:** Methodology, Visualization. **Jianqiao Han:** Validation, Writing – review & editing. **Wenyan Ge:** Visualization, Writing – review & editing. **Hao Chen:** Visualization, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

The authors do not have permission to share data.

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